

## Jargon Buster

- **Clock speed:** This is the speed at which your processor runs under factory settings, and is calculated in Gigahertz (GHz) or Megahertz (MHz). The clock speed is generally obtained from the product of FSB/HTT/BCLK and CPU multiplier. Note that whether it's FSB, HTT or BCLK depends on the type of processor you own.
- **FSB (Front Side Bus):** This acts as a main bus speed for the processor, RAM, northbridge and southbridge. The FSB lies in the motherboard and has a memory controller alongside it.
- **HTT (Hyper Transport Technology):** This is the new generation of FSBs that are used by AMD processors. HTT is nothing but the FSB, excluding a memory controller. In AMD processors, the memory controller lies inside the processor, and that provides more flexibility throughout; as a result, AMD processors support multiple memory types. For example, Phenom II, Athlon II and Sempron 140 (i.e., the 45nm series of AMD processors) have multiple memory controllers within, allowing them to use either DDR2 or DDR3 at a time.
- **BCLK (Base Clock):** This is a refined FSB that has been developed for Intel processors and debuted with the Core i7 series of processors. The BCLK is nothing but AMD's HTT. The processor comes with the memory controller. Previously, all Intel processors—Pentium, Core2Duo, or Core2Quad—had the FSB, which limited their memory handling capabilities. With Core i7, Intel has now a memory controller merged within the CPU itself and with triple channel support providing humongous bandwidth.
- **CPU multiplier:** As mentioned earlier, this value when multiplied with the FSB/HTT/BCLK, gives the processor's clock speed. And since the Special Edition processors have the multiplier unlocked, they allow processor overclocking. Thus, you don't have to tinker with the FSB/HTT/BCLK if you have a Special Edition processor. However, you are even free to take the FSB/HTT/BCLK route if you want.
- **Integrated memory controller:** This determines the speed, type, channel and the amount of memory that can be installed in the system. With new processors, the memory controller is now merged into the processor. As a result, you don't have to change the processor in order to utilise newer and faster memory, if supported. For example, the new 45nm Phenom II from AMD comes with a dual memory controller allowing you to use either DDR2 or DDR3. So you can put the processor in an AM2+ board and use DDR2, or in an AM3 board to use DDR3. This gives a wide spectrum of options to users either to opt for the high-performance DDR3 or the value for money DDR2. As a result of multi-channel memory, motherboard manufacturers release hybrid boards with both DDR2 and DDR3 slots. So all you need to do is buy one board and RAM of your choice. If, in future, you want to make the transition from DDR2 to DDR3, then you don't have to buy a new motherboard. This contrasts with processors like Intel Core2Duo or the Pentium series, which had the memory controller on board. Thus, in order to get a higher performance RAM, you need to change the motherboard, which adds to costs.
- **CPU core voltage (VCORE):** In order to gain a stable clock speed (which works 24x7 under a full load, without any problem) we need to increase the voltage applied to the processor. By applying more voltage, we give more power to operate. More power generates more heat, and thus we must opt for a good after-market CPU cooler if you are aiming for higher clocks.
- **Memory speed:** Once you overclock your system by tinkering with the FSB, HTT, or BCLK, you substantially overclock other components as well. All the other components like memory, northbridge and southbridge, thrive on FSB, HTT or BCLK for their own clock speeds. So once you change the relevant BUS, you overclock the whole system. Memory is affected as well. Memory speed is determined by the product of FSB/HTT/BCLK and the memory multiplier. Note that just like the CPU multiplier, memory too has its own multiplier.

your hardware. You can check benchmarking journals to keep yourself updated with all the reviews of the new hardware flowing into the market. Whatever said and done, it is a perfect option to test your hardware.

## Amplify speed by overclocking hardware

Overclocking is like losing weight—the initial results come very fast, but the last pounds (read MHz) take some real effort.

Almost all the hardware components can be overclocked. This is due to the fact that most of the components are interlinked and work in synchronisation. Since this requires tweaking the system above the factory settings, it means increasing the power consumption, heat dissipation and many other factors. So before we get started, here are the mandatory things that need to be dealt with:

### 1. Extremising dust:

Accumulation of dust decreases

a component's life span and performance, while decreasing the heat dissipation. Dust traps the heat and does not allow it to dissipate properly. It is very important to keep the dust away. So, open the cabinet and clean all components. Cleaning reduces noise, components run cool and hence their performance increases, significantly increasing headroom for overclocking. Make sure you are properly grounded, else the static